

Allofusion: Allosteric Site Prediction Based on Language Models and Multi-Feature Fusion

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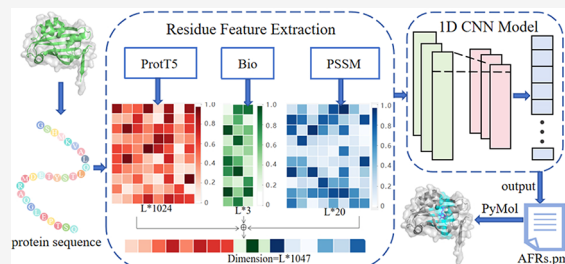


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ABSTRACT: Allosteric regulation plays a crucial role in modulating protein function and has emerged as a promising strategy in drug discovery. However, current computational methods often rely on static structures or single-modality features, limiting their ability to identify allosteric sites that are transient, cryptic, or located outside conventional pockets. Here, we propose AlloFusion, a residue-level multimodal prediction framework for accurate allosteric site prediction. AlloFusion integrates the embedding representations from a pretrained protein language model, biochemical properties of residues, and evolutionary profiles derived from position-specific scoring matrices. By leveraging these diverse features, AlloFusion effectively classifies allosteric site-forming residues (AFRs) and nonallosteric residues (FRs) in protein sequences while localizing allosteric sites. On the ASD2023 data set, AlloFusion outperforms mainstream methods in specificity, precision, F1-score, and AUC. Furthermore, on the independent test set D24, AlloFusion correctly predicts 23 out of 24 allosteric sites, significantly improving prediction accuracy. The comprehensive results demonstrate that AlloFusion is a promising method for allosteric site prediction. The source code of AlloFusion is available at <https://github.com/hjb-001/AlloFusion>.



1. INTRODUCTION

Accurate identification of allosteric sites is essential for understanding allosteric mechanisms and accelerating the development of allosteric drugs. In recent years, extensive investigations have revealed the widespread presence of allosteric regulation across various physiological and pathological processes.^{1–4} Allosteric regulation is an inherent characteristic of the protein conformational landscape, relying on specific regions within the protein molecule known as allosteric sites.⁵ Unlike orthosteric drugs, which directly bind to the active site to alter protein activity, allosteric drugs interact with allosteric sites, inducing conformational changes that indirectly regulate protein function. This mechanism enables allosteric drugs to finely tune protein activity without competing with endogenous substrates, offering distinct advantages in reducing side effects and enhancing selectivity.^{6,7} Consequently, allosteric drugs have emerged as a highly promising direction in drug research.

However, in many cases, the location of allosteric sites is unknown, making their identification a key challenge in allosteric drug discovery.⁸ Although high-resolution experimental methods such as X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and cryo-electron microscopy (cryo-EM) can identify allosteric sites, these methods are expensive and time-consuming.⁹ Consequently, various computational approaches have been developed in recent years to improve the efficiency of allosteric site identification. Existing computational methods can be broadly categorized into sequence-based and structure-based ap-

proaches.¹⁰ Sequence-based methods leverage the conservation and coevolutionary patterns of residues involved in allosteric regulation. For example, statistical coupling analysis (SCA) and related methods identify potential allosteric regulatory mechanisms by analyzing coevolving residue networks within protein families.^{11,12} Structure-based methods rely on three-dimensional (3D) protein structures, combining pocket detection algorithms (such as FPocket^{13,14} and DoGSiteScorer¹⁵ with machine learning models to identify potential allosteric sites. For instance, AlloSite¹⁶ utilizes 19 physicochemical pocket features provided by FPocket and a support vector machine (SVM) model to assign an “allosteric probability” to each pocket. Similarly, the PASSer series (including PASSer,¹⁷ PASSer 2.0,¹⁸ and PASSer Rank¹⁹) employ different machine learning algorithms following a similar approach to predict allosteric sites. Beyond static structural features, residue dynamics and other biochemical properties (e.g., residue hydrophobicity and solvent-accessible surface area; see section 2.3.3) also play crucial roles in allosteric site identification.^{20,21} For example, AllosES²⁰ predicts allosteric sites by integrating residue dynamics,

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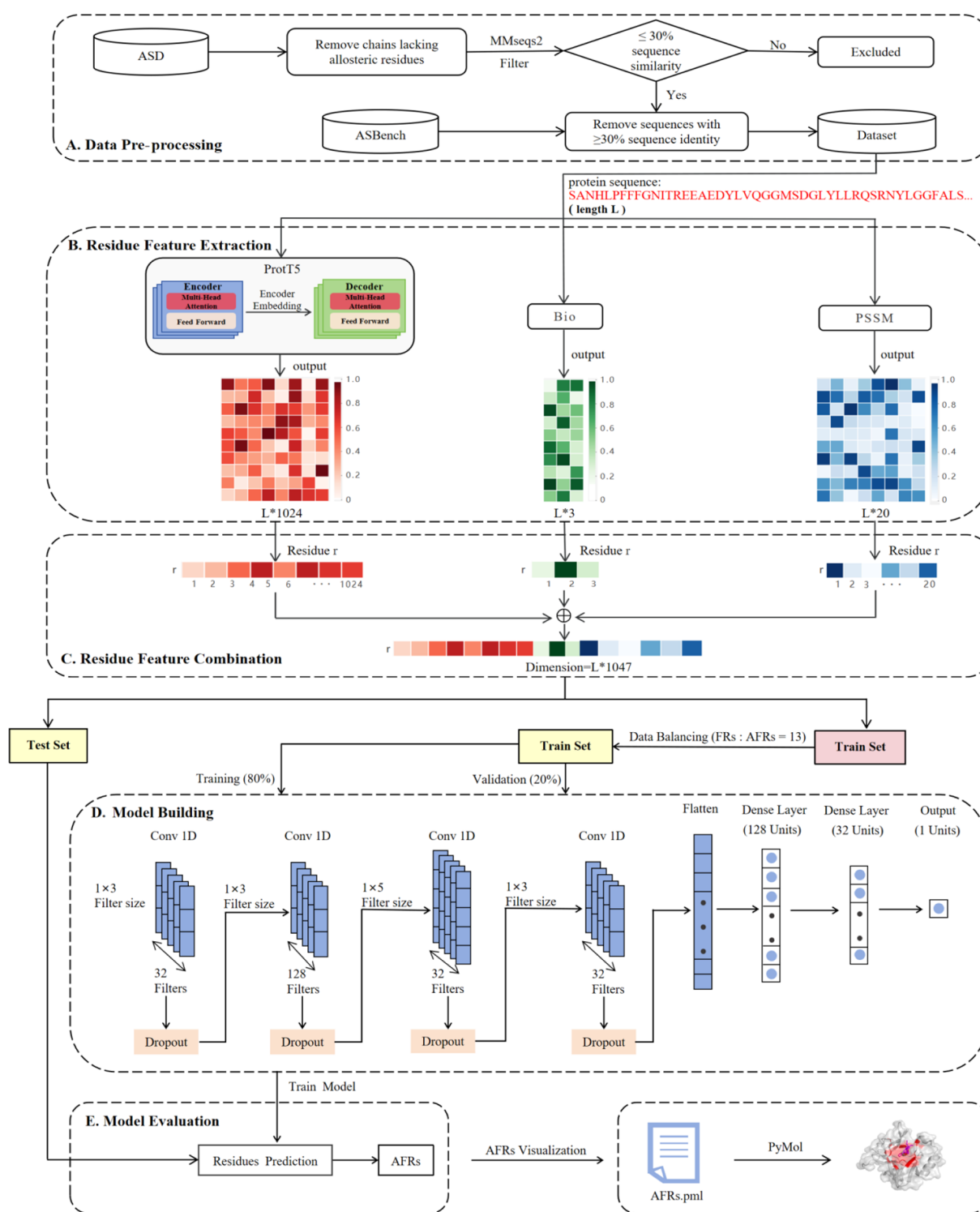


Figure 1. Workflow of AlloFusion. (A) Select protein chains from the ASD data set that exhibit less than 30% sequence identity. (B) Derive three types of residue-level features: ProtT5 embeddings, biochemical features, and PSSM profiles. (C) Concatenate the Embedding, Bio, and PSSM features for each residue to create comprehensive input vectors for the downstream model. (D) A 1D CNN is trained to classify residues as AFRs or FRs, using an undersampling strategy to address class imbalance (see section 3.1). The training set is further split into 80% for model training and 20% for validation. (E) Apply the trained model to the test set to predict AFRs and identify potential allosteric sites.

secondary structure features, and spatial neighbor-based evolutionary information.

Despite these advances, accurately predicting allosteric sites remains a formidable challenge due to their structural and evolutionary diversity.^{10,22,23} Many allosteric sites are cryptic, buried, or located on flat surfaces that are difficult to detect

through pocket-based methods. Notably, it has been reported that approximately 18.3% of allosteric site-forming residues (AFRs) lie outside detectable pocket regions, increasing the risk of missing critical regulatory sites.²⁴ These limitations underscore the need for more robust and generalizable models.

Protein representation learning based on deep learning, particularly protein language models (pLMs), offers a promising direction for addressing these challenges. In recent years, pLMs have been widely applied to various biological tasks.^{25,26} These models are pretrained on large-scale protein sequences to capture coevolutionary, structural, and functional information encoded in amino acid sequences. Since allosteric regulation is an inherent characteristic of protein structural ensembles and protein structure is implicitly encoded within amino acid sequences, pLMs pretrained on vast protein data sets possess rich internal representations.^{27–29} These representations effectively capture structural and functional relationships within sequences, making pLMs particularly well-suited for identifying residues involved in allosteric regulation. Compared to traditional sequence-based approaches, pLMs capture relationships that extend beyond simple conservation or coevolutionary patterns, offering a significant advantage. Additionally, their sequence-based nature eliminates reliance on protein structure, addressing the limitations of structure-based methods.

Nonetheless, pLMs have not yet been fully exploited for allosteric site prediction. For example, DeepAllo²⁸ integrates ProtBERT embeddings with pocket features, but remains limited by its reliance on predefined pocket detection. More recently, Allo–Allo³⁰ explores a data-efficient strategy by leveraging pretrained pLM embeddings to predict allosteric sites. Mechanistically, allosteric effects are typically driven by a small subset of critical residues that directly mediate ligand binding and functional modulation, whereas others primarily contribute to the overall conformational stability of the protein.^{12,31,32} Therefore, accurately identifying these key residues is essential for enhancing predictive performance.

Convolutional neural networks (CNNs), recognized for their efficacy in automated feature extraction and complex pattern recognition, have been extensively utilized in protein sequence modeling.^{33,34} These networks are particularly adept at managing residue-level spatial dependencies and identifying local motifs. The intricate nature of allosteric mechanisms indicates that no single type of feature can adequately encapsulate the determinants of functional modulation. Indeed, the current trend in allosteric site prediction is multifeature fusion, aiming to achieve more comprehensive and accurate prediction results. Therefore, integrating CNNs with multimodal residue-level features that combine sequence, structural, and evolutionary information presents a promising strategy for accurate allosteric site prediction.

In this study, we introduce AlloFusion, a novel residue-level prediction framework that synergistically integrates protein sequence embeddings from pLMs with CNNs. AlloFusion incorporates embeddings from the ProtT5 protein language model, the biochemical properties of residues, and position-specific scoring matrices generated from multiple sequence alignments. This comprehensive approach effectively represents residue-level information within proteins. AlloFusion employs a 1D convolutional neural network (1D CNN) to distinguish AFRs from FRs in protein sequences for residue classification. AlloFusion effectively predicts AFRs and identifies potential allosteric sites through multimodal deep feature fusion, providing a reliable computational tool to facilitate allosteric drug discovery.

2. MATERIALS AND METHODS

2.1. Overview of AlloFusion. This study introduces a framework, AlloFusion, for predicting AFRs and subsequently identifying allosteric sites. AlloFusion consists of two major components: the residue feature extraction and the residue classification. In the residue feature extraction module, three types of residue features are extracted: (i) pretrained ProtT5 embedding features (Embedding), (ii) biochemical features of residues (Bio), (iii) evolutionary information represented by position-specific scoring matrices (PSSM). Specifically, amino acid residue sequences from the benchmark data set are processed using dedicated tools to extract Embedding, PSSM, and Bio features (Figure 1A). For each residue, these features are encoded as a 1×1024 -dimensional embedding vector, a 1×20 -dimensional PSSM vector, and a 1×3 -dimensional Bio vector. These vectors are concatenated to produce a 1×1047 -dimensional residue-level feature vector (Figure 1B,C). The classification module employs a 1D CNN to assign each residue as either AFR or FR based on the integrated features (Figure 1D). During the model prediction phase (Figure 1E), for a given protein chain input, AlloFusion outputs AFRs, which can be further used to identify and locate allosteric sites. The predicted AFRs are exported in.pml format and can be visualized using PyMol³⁵ to highlight the predicted allosteric sites on the 3D protein structure.

2.2. Data Sets. The data set used in this study is obtained from the latest version (V5.0, 2023) of the Allosteric Database (ASD).³⁶ This version contains 3,102 entries of allosteric sites, each involving a distinct protein and its corresponding modulator. Although the ASD database provides a large amount of data, it is crucial to acknowledge that redundancy exists within protein sequences contained in the database.³⁷ Protein sequences were clustered using MMseqs2³⁸ with a sequence similarity threshold of less than 30%¹⁶ to ensure data quality and sequence diversity. Subsequently, a representative protein was selected from each cluster. After preprocessing, we identified 408 proteins from the original ASD data set. We further incorporated 147 structurally diverse proteins from the ASBench dataset,³⁹ a widely used benchmark derived from earlier versions of ASD. After removing duplicate entries and filtering out protein sequences with more than 30% similarity, a benchmark data set of 462 nonredundant proteins was constructed. These 462 proteins were divided into a training set (trainSet1, TR372) and a test set (testSet1, TE90) in an 8:2 ratio. Furthermore, to comprehensively evaluate the effectiveness of our model in predicting allosteric sites, we used an external test set from the AllositePro study, which includes 24 allosteric proteins. This data set is denoted as D24 (see Table S1 for details). Notably, the proteins in D24 are not included in our model's training data set. Since the full training data sets of AllositePro and PASSer are not publicly available, we cannot definitively exclude the possibility of overlap between their training data and our D24 or TE90 test sets.

AFRs are defined as specific residues within a protein participating in allosteric regulation by interacting with allosteric modulators.²⁴ In this study, amino acid residues located within 5 Å of the bound allosteric modulator in the resolved protein–ligand complex structure were labeled as AFRs. All remaining residues were considered nonallosteric residues (FRs). The final benchmark data set comprised 144,465 FRs (labeled as 0 in the experiments) and 5,239 AFRs

(labeled as 1 in the experiments). Table 1 presents the distribution of AFRs and FRs across the data set.

Table 1. Summary of the Benchmark Dataset

	trainSet1 (TR372)	testSet1 (TE90)
Number of proteins	372	90
Number of residues	149704	32284
Number of AFRs	5239	1226
Number of FRs	144465	31058

2.3. Feature Extraction. The features used in this study include embeddings from a pretrained pLM, evolutionary information derived from protein sequences using the PSI-BLAST tool,⁴⁰ and the biochemical properties of residues. As shown in Figure 1B, the feature extraction step processes input protein sequences to generate Embedding, PSSM, and Bio features, providing a comprehensive multidimensional representation of each residue. The following sections provide a detailed explanation of the feature extraction process and its biological significance.

2.3.1. Embedding Features from Protein Language Model. In protein research, traditional protein representation methods struggle to effectively capture the deep semantic information within amino acid residue sequences, thereby limiting their performance in complex tasks.²⁵ Transformer-based language models⁴¹ have proven particularly effective in protein-related tasks. Their transfer learning capability allows the application of knowledge gained from data-rich environments to tasks with limited data. The core mechanism of the Transformer is multihead attention, which utilizes Query (Q), Key (K), and Value (V) vectors to capture complex dependencies among residues in a sequence, generating rich and diverse feature representations. The following equations define the multihead attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (2)$$

$$\begin{aligned} \text{MultiHead}(Q, K, V) \\ = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_n)W^M \end{aligned} \quad (3)$$

where d_k is the dimension of K, and W_i^Q , W_i^K , W_i^V , W^M are weight matrices.

This study selected the ProtT5 model⁴² due to its outstanding performance in various protein-related tasks, such as function prediction and sequence alignment. ProtT5 was pretrained on the BFD database and subsequently fine-tuned on UniRef50,⁴³ enabling it to acquire generalized semantic features from extensive protein sequences. We fine-tuned a portion of ProtT5's parameters using Low-Rank Adaptation (LoRA)⁴⁴ to enhance the learning of allosteric residue features. A comprehensive list of hyperparameters used for fine-tuning, including batch size, learning rate, optimizer settings, number of epochs, early stopping criteria, and LoRA configuration, is provided in Supplementary Table S2. Specifically, low-rank matrices were injected into the self-attention and encoder-decoder attention layers, thereby restricting the scope of fine-tuning to improve training efficiency. The fine-tuning process was conducted on the

training set (TR372), utilizing prediction accuracy as the primary evaluation metric to ensure that LoRA fine-tuning effectively enhanced the model's ability to identify allosteric residues.

From the final hidden state of the fine-tuned ProtT5 model, we extracted embedding representations for individual residues, resulting in a feature matrix of dimensions (n , 1024), where n represents the length of the protein sequence and 1024 denotes the dimensionality of each residue's feature vector. The resulting matrix effectively captures local dependencies and long-range interactions among amino acid residues in the protein sequence by leveraging the Transformer attention mechanism.

2.3.2. Position-Specific Scoring Matrix with Sliding Window. Sequence conservation plays a crucial role in protein structure and function prediction. In this study, we utilized the PSI-BLAST⁴⁰ to extract evolutionary information for residues. Previous research has demonstrated that evolutionary information derived from varying sliding window lengths performs well in predicting protein–protein interaction (PPI) sites.⁴⁵ Specifically, we employed PSI-BLAST to conduct three iterative searches against a nonredundant protein sequence database for each protein sequence, setting the E -value threshold to 0.001 to generate a standard PSSM file. This PSSM file encapsulates evolutionary information for each residue position as a 20-dimensional vector, ultimately resulting in a matrix of dimensions $20 \times n$, where n represents the length of the protein sequence. Then, a sliding window was applied to the PSSM matrix to extract local evolutionary information for each residue. Finally, a sigmoid function (eq 4) was used to normalize the score x for each residue, mapping it to the range $[0, 1]$ to ensure uniform data scaling and compatibility with subsequent deep learning models.

$$\bar{x} = \frac{1}{1 + e^{-x}} \quad (4)$$

2.3.3. Biochemical Characteristics. To comprehensively characterize AFRs, we selected three biochemical features: residue hydrophobicity (bio_1), solvent-accessible surface area (bio_2), and dynamic properties (bio_3). Hydrophobicity and solvent-accessible surface area reflect the chemical properties and spatial exposure of residues, which are valuable for the identification of AFRs.^{46,47} The values of bio_1 and bio_2 are assigned based on standard values for the 20 canonical amino acids. The dynamic properties of residues are computed using the Elastic Network Model-based Essential Site Scanning Analysis (ESSA) method.⁴⁸ ESSA assigns a z-score to each residue, quantifying its impact on global protein dynamics upon ligand binding. Collectively, these features capture both local biochemical properties and ligand-induced dynamic effects at allosteric sites.

Nonstandard amino acid residues' corresponding physicochemical scores are set to 0. All residue features are consolidated into a single array residue_{type} and normalized using eq 5.

$$\text{residue}_{type} = \frac{[bio_1, bio_2, bio_3] - \text{residue}_{min}}{\text{residue}_{max} - \text{residue}_{min}} \quad (5)$$

where residue_{max} , residue_{min} represent the maximum and minimum physicochemical scores of residues in the protein sequence.

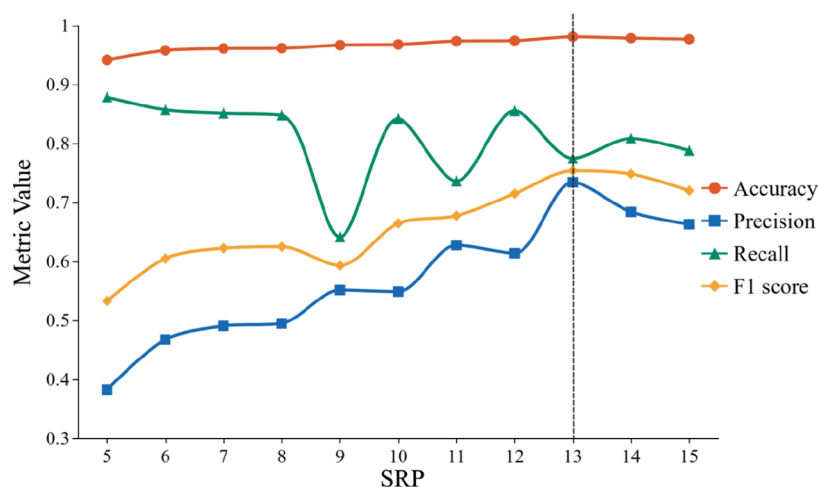


Figure 2. Model performance on the validation set under different SRP values. An SRP of 13 yields the highest F1-score, indicating an optimal trade-off between precision and recall.

In summary, we integrated embeddings from the pretrained ProtT5 model, PSSM profiles, and biochemical properties to construct a residue-level feature matrix of shape $(n, 1047)$, where n represents the protein sequence length and 1047 denotes the feature dimension for each residue. This comprehensive representation captures sequence, evolutionary, and biochemical information, providing rich input for downstream classification tasks.

2.4. Building Deep Learning Model. During the model training phase (Figure 1D), we constructed a 1D CNN architecture for residue classification using the TensorFlow framework.⁴⁹ The model was designed to process residue-level features and predict the probability of each residue being an AFR. We used the Keras Tuner library with Bayesian optimization⁵⁰ to optimize the model architecture and training parameters. All hyperparameter values and search ranges used in the tuning process are provided in Supplementary Table S3. The final architecture consists of four convolutional layers and two fully connected layers. The first convolutional layer contains 32 filters of size 3, the second layer has 128 filters of size 3, the third layer has 32 filters of size 5, and the fourth layer has 32 filters of size 3. The stride for each convolutional layer is set to 1, while “same padding” is applied to ensure that the output dimensions correspond precisely with those of the input. In terms of fully connected layers, the first and second layers contain 128 and 32 neurons, respectively. The output layer consists of a single neuron, using the sigmoid activation function to output the probability of a residue being an allosteric residue. ReLU activation functions are employed throughout all convolutional and fully connected layers to enhance the model’s capacity for nonlinear representation. Dropout layers are inserted after each convolutional layer to prevent overfitting, randomly discarding a portion of neurons to improve the model’s generalization ability.

We train the model using 80% of the training set, with the remaining 20% reserved for validation. The model is optimized using the Adam optimizer with a learning rate of 1×10^{-5} , and binary cross-entropy is used as the loss function. Model performance is primarily evaluated using the Area Under the Curve (AUC) metric. An early stopping strategy with a patience of 5 is employed during training to prevent overfitting. The maximum number of epochs is set to a hyperparameter of 100. Training stops in two cases: when the

training epoch reaches 100 or when the AUC value of the validation set does not improve in 5 consecutive epochs. Upon termination of training, the model saves the parameters from the epoch with the highest validation AUC. A callback function ensures that the model weights with the best validation performance are retained for subsequent prediction or evaluation.

2.5. Evaluation Metrics. Predicting residues constituting allosteric sites on proteins is a binary classification problem. Given the highly imbalanced distribution of positive and negative samples in the test set, selecting appropriate metrics for evaluating the model’s classification performance is essential. Therefore, in line with previous studies, we employ sensitivity (SEN), specificity (SPE), precision (PRE), F1-score (F1), and Matthew’s correlation coefficient (MCC) as evaluation metrics to evaluate the performance of our method. The formulas for these evaluation metrics are given below.

$$\text{SEN} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

$$\text{SPE} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (7)$$

$$\text{PRE} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (8)$$

$$\text{F1} = \frac{2 \times \text{SEN} \times \text{PRE}}{\text{SEN} + \text{PRE}} \quad (9)$$

$$\text{MCC} = \frac{(\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN})}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}} \quad (10)$$

where TP represents true positives (the number of actual AFRs correctly classified by the model), TN represents true negatives (the number of FRs correctly classified by the model), FP represents false positives (the number of actual FRs incorrectly classified by the model), and FN represents false negatives (the number of actual AFRs incorrectly classified by the model). In addition to the indicators above, we also use the AUC metric, which is calculated based on classification probability values and is independent of threshold settings.

3. RESULTS

3.1. Handling Data Imbalance and Optimizing Sample Proportions. In the data set used, the ratio of AFRs to FRs is approximately 1:27, which introduces a significant class imbalance and presents challenges for model training. To alleviate this issue, we employed undersampling to adjust the number of FRs in the training data set. Specifically, we introduced a sampling ratio parameter (SRP), defined as

$$\text{SRP} = \frac{N(\text{FRs})}{N(\text{AFRs})} \quad (11)$$

Where $N(\text{FRs})$ denotes the number of nonallosteric residues, $N(\text{AFRs})$ denotes the number of allosteric residues. To determine the optimal SRP value, we conducted experiments across a range from 5 to 15 and evaluated model performance on the validation set. As illustrated in Figure 2, evaluation results for different SRP values are displayed. The results indicate that when set at an SRP value of 13, the model achieves its highest F1-score, with other metrics such as accuracy and precision also performing well. Therefore, the SRP value of 13 was selected as the optimal sampling parameter for the final model.

To further assess the model's robustness under imbalanced conditions, we computed the area under the precision-recall curve (AUPRC) on the TE90 test set with $\text{SRP} = 13$. As shown in Figure 3, the model achieved an AUPRC score of

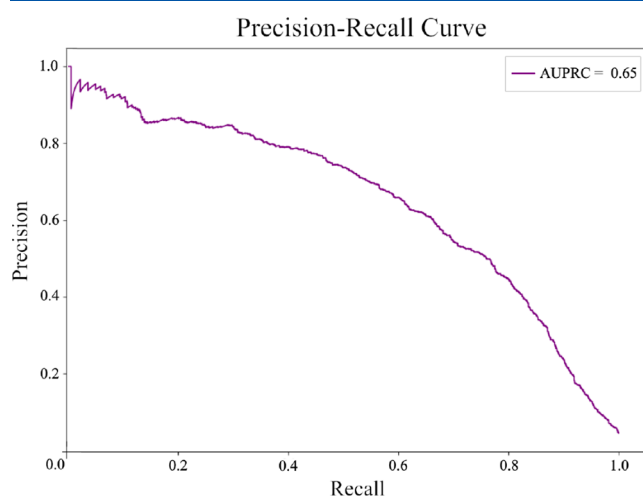


Figure 3. Precision-Recall curve of the proposed model on the TE90 test set ($\text{SRP} = 13$). The model achieves an AUPRC of 0.65, indicating strong performance under class imbalance.

0.65, substantially outperforming the random baseline. These results confirm that the proposed sampling strategy effectively mitigates class imbalance and enhances model reliability in allosteric residue prediction.

3.2. Effect of LoRA-Based Fine-Tuning on Embedding Features. To assess the necessity and effectiveness of fine-tuning the protein language model, we performed a comparative analysis utilizing embeddings from both the original pretrained ProtT5 model and the LoRA fine-tuned ProtT5 model. In both scenarios, these embeddings were input into an identical CNN architecture for predicting allosteric residues. As illustrated in Table 2, the model trained with LoRA fine-tuned embeddings demonstrated a significant performance enhancement compared to that using raw

Table 2. Performance Comparison between Models Using Pretrained and LoRA Fine-Tuned ProtT5 Embeddings

Embedding Source	SEN	SPE	PRE	F1	AUC
Pretrained embedding	0.127	0.987	0.285	0.175	0.772
LoRA fine-tuned embedding	0.181	0.997	0.710	0.289	0.902

embeddings. Specifically, there was an improvement in F1-score from 0.175 to 0.289, precision increased from 0.285 to 0.710, and AUC rose from 0.772 to 0.902. These results indicate that fine-tuning effectively enhances embedding representations by tailoring them to the specific task of allosteric site identification, thereby substantially augmenting the discriminative power of the downstream predictor.

3.3. Insights into Residue Features. To systematically evaluate the impact of different feature combinations on AFRs prediction performance, we trained and tested models using the following feature sets under identical hyperparameter settings: (i) PSSM, (ii) PSSM + Bio, (iii) Embedding, (iv) Embedding + Bio, (v) Embedding + PSSM, and (vi) Embedding + PSSM + Bio (AlloFusion). As shown in Figure 4A, the ROC curves compare the performance of different feature combinations. Notably, compared to the PSSM feature set ($\text{AUC} = 0.695$) and the PSSM + Bio combination ($\text{AUC} = 0.766$), the ProtT5 embedding feature set achieved a significantly higher AUC value of 0.902. Since the structure and function of proteins are encoded by their sequences^{51,52} this implies that embeddings derived from pretrained pLMs effectively capture information inherent in the protein sequences that are pertinent to both structural and functional aspects. Given the intrinsic conformational plasticity of allosteric sites^{53,54} such sequence-derived structural descriptors significantly improve allosteric site prediction accuracy. Integrating PSSM or Bio features with ProtT5 embeddings further enhanced model performance. The combination of Embedding + Bio features yielded the most substantial improvement ($\text{AUC} = 0.950$), while the Embedding + PSSM combination slightly increased ($\text{AUC} = 0.937$). Furthermore, the combination of Embedding + PSSM + Bio achieved the highest overall AUC of 0.953. The comparison of these feature combinations across evaluation metrics (SEN, PRE, MCC, F1, and AUC) is shown in Figure 4B.

These results indicate that the ProtT5 embedding features capture sequence- and structure-related information far beyond traditional features, making them crucial for allosteric residue prediction. The PSSM features, serving as traditional conservation indicators, effectively complement the pLM embeddings, while biochemical properties provide additional perspectives for prediction. These findings validate the efficacy of integrating diverse features into model construction and offer valuable insights for optimizing frameworks for allosteric site prediction. The superior performance of the AlloFusion model highlights the potential of multimodal feature fusion strategies in enhancing the understanding and prediction of protein AFRs.

3.4. Comparison with Other Allosteric Site Prediction Methods. To evaluate the performance of the AlloFusion model, we conducted a comparative analysis with existing methods (PASSer Ensemble, PASSer Automl, and PASSer Rank) using the TE90 test set. We focused exclusively on their top-ranked predicted pocket for pocket-based models, representing the most probable allosteric site identified by

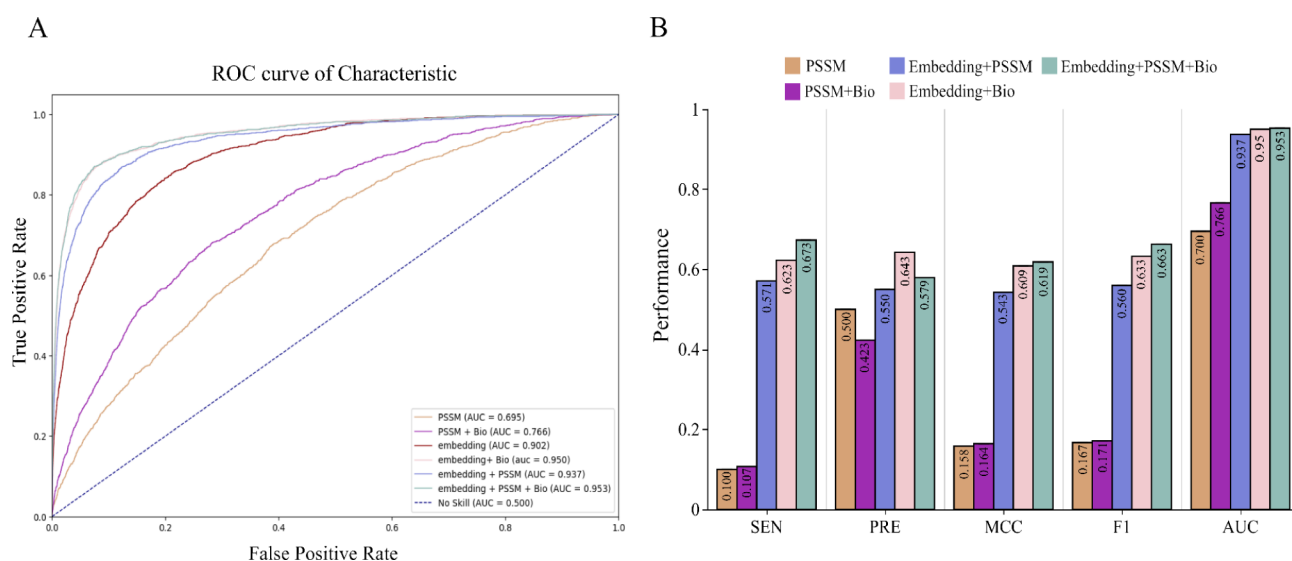


Figure 4. Impact of feature combinations on model performance (Ablation Study). (A) Comparison of ROC curves on the test set. (B) Evaluation of SEN, PRE, MCC, F1, and AUC on the test set.

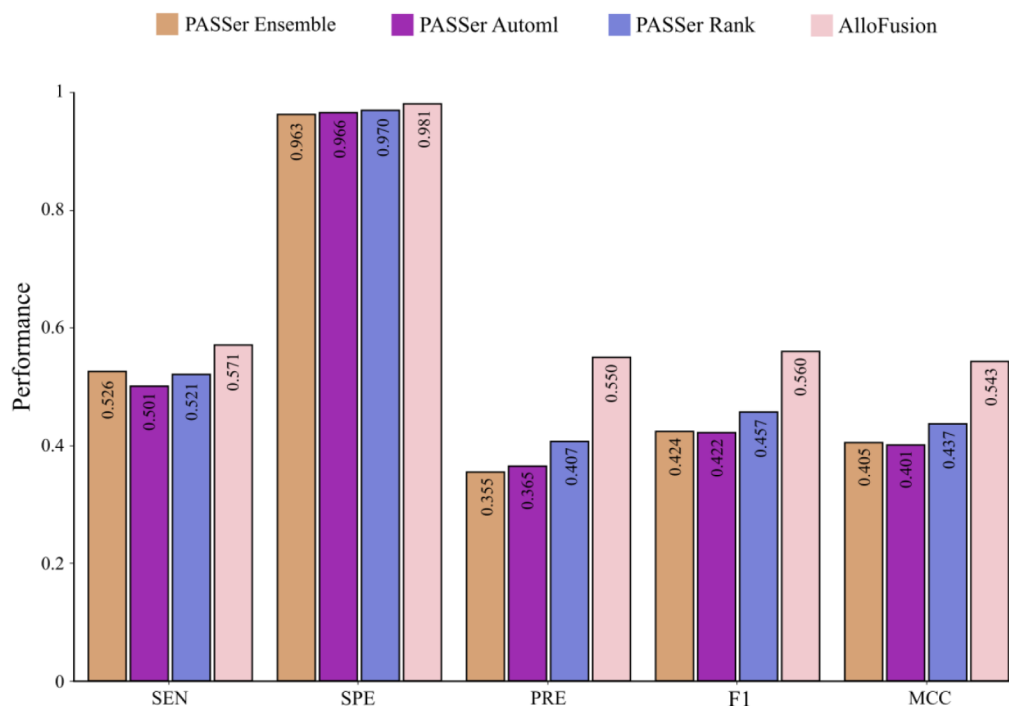


Figure 5. Comparison of AFRs prediction performance on the TE90 test set among AlloFusion, PASSer Ensemble, PASSer Automl, and PASSer Rank.

these models. The residues constituting this top-ranked pocket were extracted as their predicted AFRs to ensure fairness and consistency in our comparisons. We used SEN, SPE, PRE, F1, and MCC to assess the performance of different methods. Figure 5 illustrates the performance of AlloFusion compared to other methods on the TE90 test set. As demonstrated, AlloFusion achieved SEN, SPE, PRE, F1, and MCC values of 0.571, 0.981, 0.550, 0.560, and 0.543, respectively. Notably, AlloFusion exhibited improvements across all metrics when compared to alternative methods. For instance, in comparison with the second-best method PASSer Rank (PRE = 0.407, F1 = 0.457), AlloFusion improved PRE by 35% and F1 by 22.5%. The superior prediction performance for AFRs observed with

AlloFusion may be attributed to two primary factors. First, the protein language model embeddings utilized by AlloFusion effectively capture inter-relationships among different amino acid residues within proteins. Second, the synergistic effects derived from multisource features enable the comprehensive characterization of each amino acid residue, enhancing its ability to identify AFRs and ensuring more accurate predictions.

Furthermore, to further validate the effectiveness of AlloFusion in predicting allosteric sites, we compared our method against AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl, and DeepAllo using the test set D24. In AlloFusion, residues with predicted probabilities ≥ 0.5 were

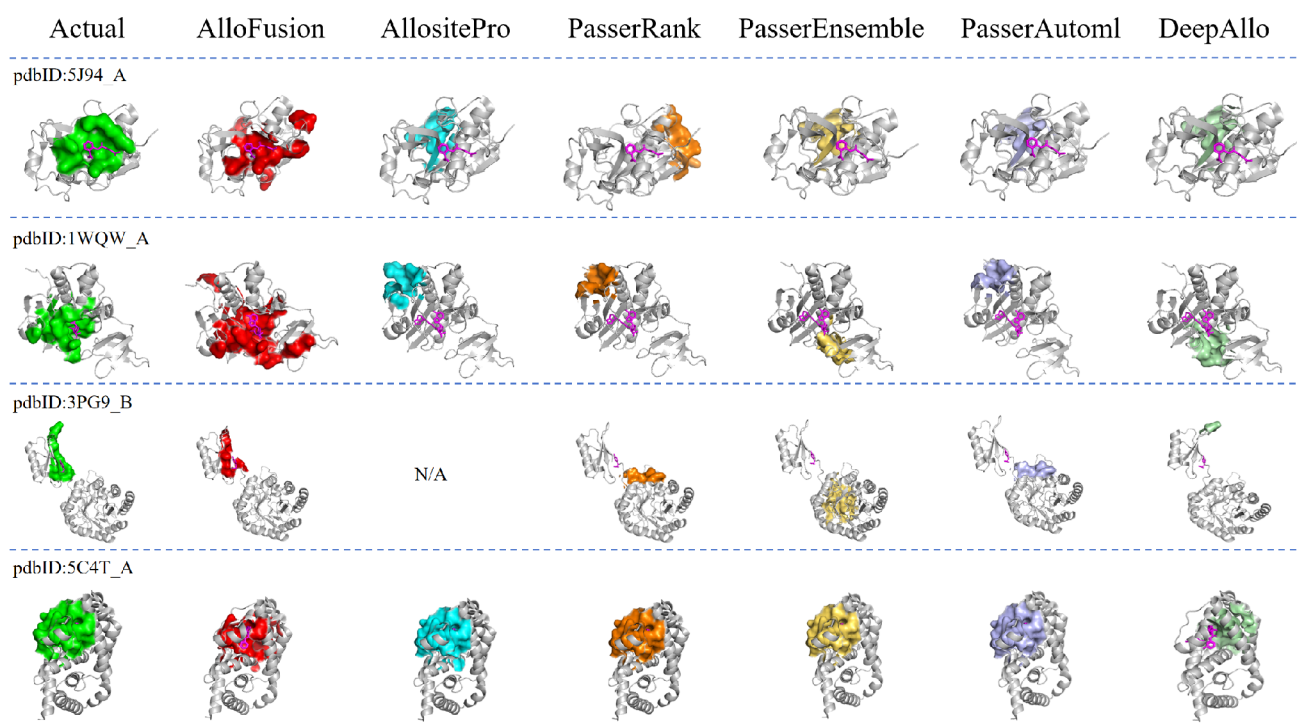


Figure 6. 3D visualization of AFRs predicted by different models for selected proteins. Each row shows: actual AFRs (green), followed by predictions from AlloFusion (red), AllositePro (cyan), PASSer Rank (orange), PASSer Ensemble (light yellow), PASSer Automl (light blue), and DeepAllo (light green). "N/A" indicates that the method does not predict any allosteric site, while gray represents the cartoon structure of the protein.

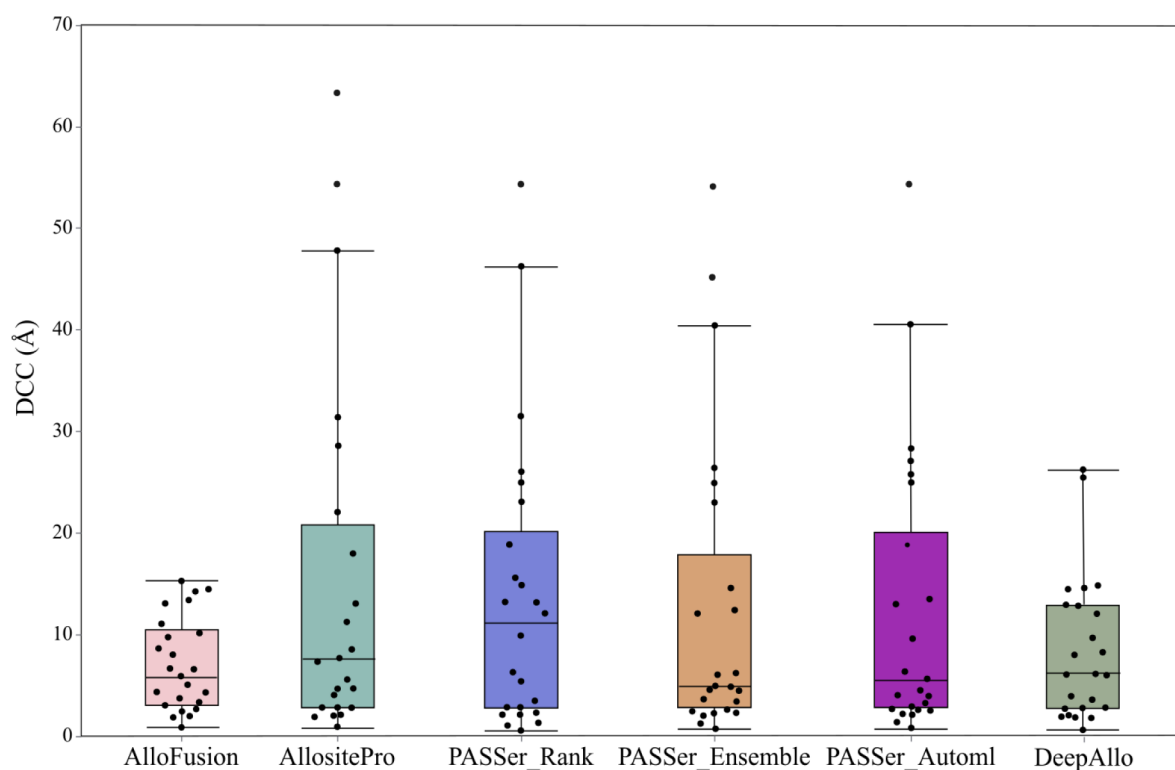


Figure 7. Box plot of DCC value distributions for AlloFusion, AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl, and DeepAllo on the test set D24.

classified as AFRs. An allosteric site was then defined as the spatial cluster formed by these predicted AFRs. As in the TE90 test set experiments, we focused solely on the top-ranked

pocket residues as their predicted AFRs for pocket-based models. We visualized the allosteric sites composed of AFRs predicted by these models in D24 using PyMol (see Figure S1

for details). The 3D plots of predicted AFRs for selected proteins in the D24 generated by different models are presented in Figure 6. For example, in chain A of protein 5J94,⁵⁵ AlloFusion accurately identified key residues (Asn199, Lys122, Arg123, and Ala1) that interact with the modulator 1XF, while the other methods failed to predict these critical residues. Notably, for proteins where pocket-based methods completely missed the true allosteric sites (e.g., Chain A of 5J94 and Chain B of 3PG9), subsequent analysis using FPocket revealed an absence of detectable pockets near the actual binding regions (Figure S2). This limitation of pocket-centric approaches highlights their inherent dependency on predefined geometric criteria, which may overlook spatially dispersed or conformationally dynamic allosteric residues. In contrast, AlloFusion is less dependent on pocket features and is capable of detecting allosteric sites that are difficult to detect using pocket-based methods (see Figure S3 for details).

The Distance Center Center (DCC) metric⁵⁶ was employed to quantify the spatial distance between the predicted allosteric sites and the actual allosteric sites. DCC measures the Euclidean distance between the centroid of the predicted AFRs and the centroid of the actual binding site residues, calculated as follows:

$$\text{DCC} = \sqrt{(x_{\text{pred}} - x_{\text{true}})^2 + (y_{\text{pred}} - y_{\text{true}})^2 + (z_{\text{pred}} - z_{\text{true}})^2} \quad (12)$$

where $(x_{\text{pred}}, y_{\text{pred}}, z_{\text{pred}})$, $(x_{\text{true}}, y_{\text{true}}, z_{\text{true}})$ represent the centroid coordinates of the predicted and actual AFRs, respectively.

As shown in Figure 7, the boxplot compares the DCC value distributions across all models on the D24 (see Table S4 for details). Experimental results indicate that although the prediction results varied among different methods, AlloFusion consistently achieved a DCC value below 10 Å in D24. These results demonstrate that AlloFusion's predicted allosteric sites were spatially closer to the actual allosteric sites. For example, on chain A of protein 1WQW, AlloFusion achieved a DCC of 2.6 Å, whereas the best DCC value among the other five methods was 12.69 Å, significantly outperforming other methods. Similarly, on chain B of 3PG9, where the allosteric site is located on a flat surface region, AlloFusion achieved a DCC of 2.59 Å. In contrast, AllositePro could not identify any allosteric site, and the three PASSer models failed to predict the site correctly. The failure of specific models to predict AFRs may stem from their internal algorithms' shortcomings in identifying conformationally viable AFRs within the established structural constraints.

Finally, to evaluate the overall success rate of allosteric site identification on the test set D24, we employed the Effective Prediction Rate (EPR) metric, defined as

$$\text{EPR} = \frac{N_{\text{pred}}}{N_{\text{total}}} \quad (13)$$

where N_{pred} represents the number of proteins in D24 for which the model correctly predicted the allosteric site, and $N_{\text{total}} = 24$ is the total number of proteins in the data set. An allosteric site prediction is regarded as effective if it includes at least one residue that coincides with the actual allosteric site.⁵⁷

A comparison of spatial localization accuracy and effective prediction rate between AlloFusion and other methods on the D24 test set is summarized in Table 3. The results indicate that

Table 3. Average DCC Values (Reported as Mean $\pm$ Standard Error) and the Number of Successfully Predicted Allosteric Sites for AlloFusion, AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl and DeepAllo on the Test Set D24

Method	DCC_AVG (Å)	EPR
AlloFusion	7.16 $\pm$ 0.89	23/24
AllositePro	14.41 $\pm$ 3.66	14/24
PASSer Rank	13.39 $\pm$ 2.76	14/24
PASSer Ensemble	11.76 $\pm$ 2.94	16/24
PASSer Automl	11.28 $\pm$ 2.62	16/24
DeepAllo	8.17 $\pm$ 1.36	22/24

AlloFusion outperformed all competing methods on both key metrics. Specifically, AlloFusion achieved the lowest average DCC value of 7.16 Å (standard deviation $\pm$ 3.60), representing a reduction in DCC by 12.4% compared to the second-best method. This significant improvement suggests that its predicted allosteric sites are spatially much closer to their actual sites. Additionally, AlloFusion successfully predicted 23 out of 24 allosteric sites in D24, resulting in an EPR of 95.8%. In contrast, the prediction success rates for AllositePro, PASSer Rank, PASSer Ensemble, PASSer Automl and DeepAllo were 58.3% (14/24), 58.3% (14/24), 66.7% (16/24), 66.7% (16/24), and 91.7% (22/24), respectively, showing a statistically significant difference ($\chi^2 = 10.79$, $p = 0.029$).

3.5. Case Study. To illustrate the residue-level prediction capability of AlloFusion, we present case studies on three representative proteins (PDB IDs: 2V4Y, 3M6F, and 6BOZ) from the test set. Figure 8 presents the 3D-visualized structures of these proteins, providing an intuitive representation of the spatial distribution of their allosteric sites.

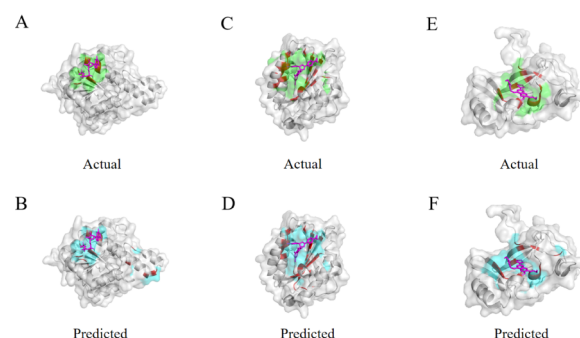


Figure 8. 3D structural visualization of three proteins (PDB IDs: 2V4Y, 3M6F, and 6BOZ) using PyMol software. The actual allosteric sites (green regions) are shown at the top (A, C, and E), while the corresponding allosteric sites predicted by our method (cyan regions) are displayed at the bottom (B, D, and F). Red highlights on the protein cartoon represent the allosteric site-forming residues. The sequence similarity between the above proteins and the proteins in the training set is less than 30%.

To quantitatively evaluate the consistency between the AFRs predicted by our model and the actual AFRs in the three proteins, we linearized their amino acid sequences and annotated both the predicted and actual AFRs (Figure 9). In this figure, green represents the actual AFRs, while red denotes AFRs predicted by AlloFusion. As illustrated in Figure 9, AlloFusion exhibits a high concordance between the expected and actual AFRs for these proteins. Specifically, our model correctly identified 6 out of 7 AFRs in 2V4Y, 16 out of 21

A	2V4Y_A: Experimental: MATNAKPVYKRILLKLSGEALQGTEGFGIDASILDRMAQEIKELVELGIQVGVVIGGGNLFVRGAGLAKAGMNRVVGDMGMLATVMNGLAMRDALHRAVYNAR LMSAIPNGVCDYSWAEISLLRNNRVVILSAGTGNPFFTTDSAACLRGIEIANVVLKATKVDGVFTADPAKDPTATMYEQLTYSEVLEKELKVMDLAAFTLARDHKLPIRVFNMN KPGALRRVVMGEKEGLITE Predicted: MATNAKPVYKRILLKLSGEALQGTEGFGIDASILDRMAQEIKELVELGIQVGVVIGGGNLFVRGAGLAKAGMNRVVGDMGMLATVMNGLAMRDALHRAVYNAR LMSAIPNGVCDYSWAEISLLRNNRVVILSAGTGNPFFTTDSAACLRGIEIANVVLKATKVDGVFTADPAKDPTATMYEQLTYSEVLEKELKVMDLAAFTLARDHKLPIRVFNMN KPGALRRVVMGEKEGLITE
B	3M6F_A: Experimental: GSHMNVDLVFLFDGSMQLQDEFQKILDFMKDVMKKLSNTSYQFAAVQFSTSYKTEFDFSDYVVRKDPDALLKHVHMLLLNTFTGAINYVATEVFREELGARPD ATKVLIIITDGEATDSGNIDAAKDIIRYIIGIKHFQTKESQETLHKFASKPASEFVKILDTFEKLDLFTLELQKKIY Predicted: GSHMNVDLVFLFDGSMQLQDEFQKILDFMKDVMKKLSNTSYQFAAVQFSTSYKTEFDFSDYVVRKDPDALLKHVHMLLLNTFTGAINYVATEVFREELGARPD ATKVLIIITDGEATDSGNIDAAKDIIRYIIGIKHFQTKESQETLHKFASKPASEFVKILDTFEKLDLFTLELQKKIY
C	6BOZ_A: Experimental: MSYYHHHHHHYDIPTTENLYFQGAMGSRKSKAELQSEERKRIDELIESGKEEGMKIDLIDGKGRGVIA TKQFSRGDFVVEYHGLDIEITDAKKREALYAQDPSTGC YMYFYQYLSKTYCVDATRETNRLGRLNHSKSGNCQTKLHDIDGVPHLLIASRDIAAGEELLYDYGDRSKASIEAHPWLKH Predicted: MSYYHHHHHHYDIPTTENLYFQGAMGSRKSKAELQSEERKRIDELIESGKEEGMKIDLIDGKGRGVIA TKQFSRGDFVVEYHGLDIEITDAKKREALYAQDPSTGC YMYFYQYLSKTYCVDATRETNRLGRLNHSKSGNCQTKLHDIDGVPHLLIASRDIAAGEELLYDYGDRSKASIEAHPWLKH

Figure 9. One-dimensional representations of the unfolded protein sequences for PDB ID: 2V4Y (A), PDB ID: 3M6F (B), and PDB ID: 6BOZ (C) corresponding to chain A are illustrated. The upper sequence for each protein highlights the actual allosteric residues (green), while the lower sequence depicts the allosteric residues predicted by our method (red).

AFRs in 3M6F, and 11 out of 15 AFRs in 6BOZ. Further supporting this analysis, Table 4 presents the detailed

Table 4. Performance of allofusion in Predicting AFRs in Three Proteins (PDB ID: 2V4Y, 3M6F, and 6BOZ)

pdBID	SEN	SPE	PRE	AUC
2V4Y	0.625	0.974	0.455	0.955
3M6F	0.810	0.932	0.607	0.917
6BOZ	0.750	0.983	0.800	0.985

performance metrics (SEN, SPE, PRE, and AUC) of AlloFusion across the three proteins. These cases, together with the results on the D24 test set (see Table S5 for details), demonstrate that AlloFusion can effectively distinguish true allosteric residues while minimizing the false positive rate. Overall, these findings confirm the model's ability to capture key allosteric residues directly from sequence data, underscoring its practical value for residue-level allosteric site prediction.

We further assessed AlloFusion's generalizability by testing it on a previously untested protein (PDB ID: 4ZSI)⁵⁸ that was not included in the training data set. Figure 10 illustrates the allosteric site region in 4ZSI as AlloFusion (cyan) predicted. As depicted in the figure, the allosteric modulator (represented by the purple stick structure) interacts with this predicted region, confirming the model's capability to accurately predict allosteric sites in unseen proteins.

4. DISCUSSION

This study demonstrated that AlloFusion can effectively predict AFRs and FRs from protein amino acid sequences. By integrating sequence-derived features from a pretrained protein language model, evolutionary conservation scores, and biochemical properties of residues, AlloFusion leverages a 1D CNN architecture to distinguish AFRs from FRs. Random undersampling was applied to mitigate the sample imbalance in the training data set, ensuring balanced learning and improving

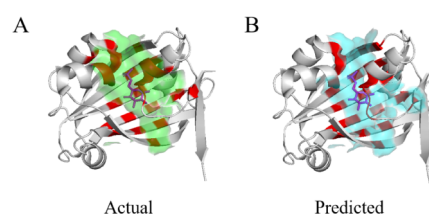


Figure 10. Prediction results of allosteric sites in PDB ID: 4ZSI by AlloFusion. In Figure A, the green region represents the actual allosteric site, while in Figure B, the cyan region indicates the predicted allosteric site. The gray represents the protein cartoon structure and red highlights on the protein cartoon represent the allosteric site-forming residues. The sequence similarity between the protein and the proteins in the training set is less than 30%.

the model's generalization performance. Unlike pocket-based methods for predicting allosteric sites, AlloFusion employs a residue-level prediction strategy that circumvents the limitations associated with pocket-centric approaches, thereby effectively identifying hidden or difficult-to-capture allosteric sites. AlloFusion achieved robust performance on the TE90 test set (SEN = 0.571, SPE = 0.981, PRE = 0.550, F1 = 0.560), highlighting its reliability in AFR identification. Additionally, on the independent test set D24, AlloFusion correctly predicted 23 out of 24 actual allosteric sites. Notably, this includes flat binding sites that pocket-based methods failed to identify (e.g., the 3PG9_B protein illustrated in Figure 6, where DCC = 2.59 Å). These results highlight the model's ability to predict conventional and unconventional allosteric sites with high spatial precision.

Furthermore, as protein language models are adept at capturing complex sequence characteristics, they have the potential to identify allosteric regulatory residues within structurally unresolved regions, such as intrinsically disordered regions. This capability facilitates the discovery of meaningful allosteric sites. If researchers can successfully identify allosteric sites in target proteins, they will be better equipped to design allosteric modulators explicitly targeting them. Consequently,

this provides a novel computational tool for advancing allosteric drug discovery.

While AlloFusion demonstrates strong performance in predicting AFRs, there is still room for further improvement and extension. First, the data set used in this study is relatively constrained and may not fully capture the diversity of residues involved in allosteric site formation across different proteins, potentially limiting model generalizability. Second, given that the ASD database may contain undiscovered allosteric sites, specific amino acid residues may have been inaccurately classified as FRs rather than AFRs. Such misclassification has the potential to introduce variability and diminish model accuracy. As more experimentally validated data become available, prediction precision is expected to improve.

To further enhance the model's generalizability and precision, we plan to incorporate newly available allosteric data sets and experimentally validated complexes as they become available. Moreover, we aim to explore spatial-aware modeling techniques, such as graph neural networks (GNNs), to better capture long-range residue interactions underlying allosteric communication. These directions are expected to further improve the interpretability and applicability of AlloFusion across a broader range of protein systems.

5. CONCLUSIONS

Identifying allosteric sites remains a fundamental challenge in structural biology with profound implications for drug discovery. Here, we present AlloFusion, a deep learning framework for residue-level prediction of allosteric sites in proteins. AlloFusion integrates embeddings from a pretrained protein language model, position-specific scoring matrices derived from multiple sequence alignments, and biochemical properties of residues. By employing a 1D CNN for classification, AlloFusion overcomes limitations associated with pocket-based methods. Experimental results on the ASD2023 data set and the independent test set D24 (EPR = 23/24) demonstrated that AlloFusion outperforms existing methods across multiple evaluation metrics (SEN, PRE, F1, AUC) while maintaining high predictive stability and accuracy. AlloFusion offers a robust and extensible computational tool for allosteric site identification. This work highlights the promise of multimodal feature integration and sequence-based deep learning in accelerating allosteric drug discovery and advancing our understanding of protein regulation.

■ ASSOCIATED CONTENT

Data Availability Statement

The source code and data sets are available at <https://github.com/hjb-001/AlloFusion>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.5c01033>.

Table S1: Independent test set D24; Table S2: Fine-Tuning Details of ProtT5; Table S3: Hyperparameter Settings of AlloFusion; Table S4: DCC value in D24; Table S5: Number and false positive rate of AFRs for each protein in D24. Figure S1: 3D representations of protein AFRs predicted by different models on the D24 data set; Figure S2: Fpocket-based binding pocket prediction; Figure S3: potential cryptic allosteric sites on protein 5J94 (PDF)

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Author Contributions

J.H. and Y.L. implemented the AlloFusion model. J.H., D.G., Y.W., and M.L. contributed to the writing, revision, and editing of the manuscript. D.G. supervised and designed the project. J.H. and Y.L. completed the modification and program optimization of the paper after revision.

Notes

The authors declare no competing financial interest.

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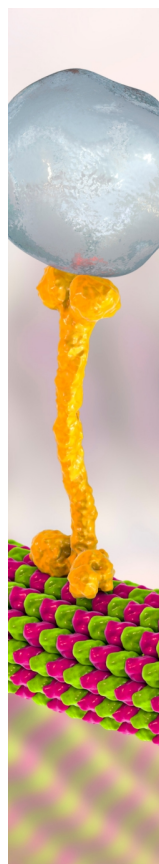
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